

Computer virus and malware are related but distinct concepts in the realm of computer security. Here are the key differences between them:

- **Nature:**
- **Computer Virus:** A computer virus is a specific type of malware. It is a self-replicating program or code that attaches itself to legitimate software or files. Viruses can spread from one computer to another when infected files are shared or transferred.
- **Malware:** Malware is a broad term that encompasses various types of malicious software. It includes viruses, as well as other malicious programs such as Trojans, worms, spyware, adware, ransomware, and more.
- **Propagation:**
- **Computer Virus:** Viruses require a host program or file to infect. They spread when the infected host is executed, and the virus code becomes part of the host, potentially infecting other files or systems.
- **Malware:** Malware, in general, does not always require a host program. Malicious software like Trojans and ransomware can operate independently without attaching themselves to other files.
- **Behavior:**
- **Computer Virus:** Viruses often have a destructive or disruptive payload. They may corrupt or delete files, slow down a computer's performance, or perform other malicious actions.
- **Malware:** Malware can have a wide range of behaviors. Some malware is designed to steal sensitive information, while others aim to control a compromised system, display unwanted advertisements, or encrypt files for ransom.
- **Propagation Mechanisms:**
- **Computer Virus:** Viruses spread primarily through infected files, email attachments, removable storage devices, or downloads from infected websites.
- **Malware:** Malware can spread through various means, including email attachments, malicious links, drive-by downloads, social engineering, and software vulnerabilities.
- **Visibility:**
- **Computer Virus:** Viruses are typically more visible because they often manifest as infected files or software that users encounter.
- **Malware:** Some forms of malware can be less visible to users, operating in the background without obvious signs of infection.
- **Detection and Removal:**
- **Computer Virus:** Antivirus software is effective in detecting and removing many types of viruses by scanning files and programs for known virus signatures.
- **Malware:** To combat malware comprehensively, a combination of security measures, including antivirus, antimalware, and anti-exploit tools, is often required due to the diversity of malware types.

The attacking model of **ransomware** typically involves the following steps:

- **Infection:** Ransomware is usually delivered to a victim's computer through malicious email attachments, infected software downloads, or by exploiting vulnerabilities in the operating system or software.
- **Execution:** Once on the victim's system, the ransomware code is executed. It encrypts the victim's files and data, making them inaccessible.
- **Ransom Note:** After encryption, ransomware displays a ransom note on the victim's screen, demanding a payment (usually in cryptocurrency) in exchange for a decryption key that can unlock the encrypted files.
- **Payment:** Victims are instructed on how to make the ransom payment, often using anonymous cryptocurrencies like Bitcoin. Payment is typically demanded within a specified timeframe.
- **Decryption Key:** After receiving the ransom payment, the attackers may provide the victim with a decryption key, which can be used to unlock the encrypted files. However, there is no guarantee that the attackers will provide the decryption key or that it will work.
- **Data Release (Optional):** In some cases, attackers may threaten to release the victim's sensitive data or files publicly if the ransom is not paid, adding another layer of coercion.